

Exercise 4 - SQL_JOINS

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Data Analytics

Brightlearn

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Question1

```
31  --Step 4: Insert data into the table
32  INSERT INTO brightlearn.sql_joins.users VALUES
33  (1, 'Nomvula', 'Johannesburg'),
34  (2, 'David', 'Cape Town'),
35  (3, 'Anele', 'Durban'),
36  (4, 'Kabelo', 'Pretoria'),
37  (5, 'Lerato', 'Port Elizabeth');
38
39  SELECT *
40  FROM brightlearn.sql_joins.users
41  ORDER BY user id
```

Add parameter

Table  

	¹ ₃ user_id	^A _C user_name	^A _C country
1	1	Nomvula	Johannesburg
2	2	David	Cape Town
3	3	Anele	Durban
4	4	Kabelo	Pretoria
5	5	Lerato	Port Elizabeth

```

73  --Question 1: Show every user who has a subscription. Match users to subscriptions.
74  SELECT
75  u.user_id,
76  u.user_name,
77  s.subscription_id,
78  s.start_date
79  FROM brightlearn.sql_joins.users u
80  INNER JOIN brightlearn.sql_joins.subscriptions s
81  ON u.user_id = s.user_id
82  ORDER BY s.subscription_id;

```

Add parameter

Table ▼ +

	¹ ₂ user_id	^A _C user_name	¹ ₂ subscription_id	start_date
1	1	Nomvula	501	2026-01-15
2	2	David	502	2026-02-01
3	1	Nomvula	503	2026-03-10
4	3	Anele	505	2026-04-05

Question 2:

```

84  --Question 2: Show every subscription with its matching plan name and monthly price
85  SELECT
86  s.subscription_id,
87  s.user_id,
88  p.plan_name,
89  p.monthly_price
90  FROM brightlearn.sql_joins.subscriptions s
91  INNER JOIN brightlearn.sql_joins.plans p
92  ON s.plan_id = p.plan_id
93  ORDER BY s.subscription_id;

```

Add parameter

Table ▼ + Q Y ⌵

	¹ ₂ subscription_id	¹ ₂ user_id	¹ ₂ plan_id	start_date
1	501	1	10	2026-01-15
2	502	2	11	2026-02-01
3	503	1	12	2026-03-10
4	504	6	11	2026-03-20
5	505	3	13	2026-04-05

Question 2:

```
108 SELECT DISTINCT
109 s.subscription_id,
110 s.user_id,
111 p.plan_name,
112 p.monthly_price
113 FROM brightlearn.sql_joins.subscriptions s
114 INNER JOIN brightlearn.sql_joins.plans p
115 ON s.plan_id = p.plan_id
116 ORDER BY s.subscription_id;
117
```

Add parameter

Table  

	^{1 2} ₃ subscription_id	^{1 2} ₃ user_id	^{A B} _C plan_name	^{1 2} ₃ monthly_price
1	501	1	Basic	79
2	502	2	Standard	129
3	503	1	Premium	199
4	504	6	Standard	129
5	505	3	Family	249

Question 3

```
121 SELECT DISTINCT
122   v.session_id,
123   v.user_id,
124   s.show_title,
125   s.genre,
126   v.watch_minutes
127 FROM brightlearn.sql_joins.viewing_sessions v
128 INNER JOIN brightlearn.sql_joins.shows s
```

Add parameter

Table ▾ +

	¹ ₂ ³ session_id	¹ ₂ ³ user_id	^A _B ^C show_title	^A _B ^C genre	¹ ₂ ³ watch_minutes
1	902	2	Tech Tales	Documentary	30
2	901	1	Comedy Hour	Comedy	45
3	903	1	Crime Time	Drama	60
4	904	7	Comedy Hour	Comedy	20
5	905	3	Wild Earth	Documentary	90

Question 4

```
133 SELECT DISTINCT
134     u.user_name,
135     u.country,
136     v.session_id,
137     v.show_id,
138     v.watch_minutes
139 FROM brightlearn.sql_joins.users u
140 INNER JOIN brightlearn.sql_joins.viewing_sessions v
141 ON u.user_id = v.user_id;
142
```

Add parameter

Table ▼ +					
	^A _C user_name	^A _C country	¹ ₃ session_id	¹ ₃ show_id	¹ ₃ watch_minutes
1	Nomvula	Johannesburg	901	701	45
2	David	Cape Town	902	703	30
3	Anele	Durban	905	705	90
4	Nomvula	Johannesburg	903	702	60

Question 5:

```
118 --question 3: Show every viewing session that has a matching show. Include the show title and
119 --genre.
120
121 SELECT DISTINCT
122 v.session_id,
123 v.user_id,
124 s.show_title,
125 s.genre,
126 v.watch_minutes
127 FROM brightlearn.sql_joins.viewing_sessions v
128 INNER JOIN brightlearn.sql_joins.shows s
```

Add parameter

	^A _C user_name	^A _C country	^A _C plan_name	¹ ₃ monthly_price	start_date
1	Nomvula	Johannesburg	Basic	79	2026-01-15
2	Nomvula	Johannesburg	Premium	199	2026-03-10
3	David	Cape Town	Standard	129	2026-02-01
4	Anele	Durban	Family	249	2026-04-05

Question 6

```
158 --Question 6: Show every user and any subscriptions they have. Users without subscriptions must still
159 --appear.
160 SELECT DISTINCT
161 u.user_id,
162 u.user_name,
163 s.subscription_id,
164 s.start_date
165 FROM brightlearn.sql_joins.users u
166 LEFT JOIN brightlearn.sql_joins.subscriptions s
167 ON u.user_id = s.user_id;
```

Add parameter

	¹ ₃ user_id	^A _C user_name	¹ ₃ subscription_id	start_date
1	1	Nomvula	501	2026-01-15
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	null	null
6	5	Lerato	null	null

Question 7

```
169 --question 7: Show every plan and the subscriptions on it. Plans with no subscribers must still
170 --appear.
171 SELECT DISTINCT
172   p.plan_id,
173   p.plan_name,
174   s.subscription_id,
175   s.user_id
176 FROM brightlearn.sql_joins.plans p
177 LEFT JOIN brightlearn.sql_joins.subscriptions s
178 ON p.plan_id = s.plan_id;
```

Add parameter

Table ▼ +					
	plan_id	plan_name	subscription_id	user_id	
1	11	Standard	504	6	
2	11	Standard	502	2	
3	10	Basic	501	1	
4	12	Premium	503	1	
5	13	Family	505	3	
6	14	Mobile	null	null	

Question 8

--Question 8:Show every show and any viewing sessions on it. Shows that were never watched must
--still appear.

```
SELECT DISTINCT  
s.show_id,  
s.show_title,  
v.session_id,  
v.watch_minutes  
FROM brightlearn.sql_joins.shows s  
LEFT JOIN brightlearn.sql_joins.viewing_sessions v  
ON s.show_id=v.show_id;
```

parameter

show_id	show_title	session_id	watch_minutes
704	Cooksing Lab	null	null
701	Comedy Hour	904	20
703	Tech Tales	902	30
702	Crime Time	903	60
706	Wild Earth	null	null
701	Comedy Hour	901	45
705	Wild Earth	905	90

Question 9

--Question 9: Show every viewing session and the user who watched it. Sessions referencing users that do not exist must still appear (with NULL user details).

```
SELECT DISTINCT
v.session_id,
v.show_id,
v.watch_minutes,
u.user_id,
u.user_name,
u.country
FROM brightlearn.sql_joins.viewing_sessions v
LEFT JOIN brightlearn.sql_joins.users u
ON v.user_id = u.user_id
```

parameter

session_id	show_id	watch_minutes	user_id	user_name	country
904	701	20	null	null	null
902	703	30	2	David	Cape Town
905	705	90	3	Anele	Durban
903	702	60	1	Nomvula	Johannesburg
901	701	45	1	Nomvula	Johannesburg

Question 10

--Question 10 Show every user, the plan they are on (if any), and the monthly price. Users without a
--subscription must still appear.

```
SELECT DISTINCT  
u.user_name,  
u.country,  
p.plan_name,  
p.monthly_price  
FROM brightlearn.sql_joins.users u  
LEFT JOIN brightlearn.sql_joins.subscriptions s  
ON u.user_id = s.user_id  
LEFT JOIN brightlearn.sql_joins.plans p  
ON s.plan_id = p.plan_id;
```

parameter

user_name	country	plan_name	monthly_price
David	Cape Town	Standard	129
Lerato	Port Elizabeth	null	null
Anele	Durban	Family	249
Kabelo	Pretoria	null	null
Nomvula	Johannesburg	Premium	199
Nomvula	Johannesburg	Basic	79

Question 11

```
219 --question 11:Show every user and every subscription, including users without subscriptions AND
220 --subscriptions referencing users that do not exist.
221
222 SELECT DISTINCT
223     u.user_id,
224     u.user_name,
225     s.subscription_id,
226     s.start_date
227 FROM brightlearn.sql_joins.users u
228 FULL OUTER JOIN brightlearn.sql_joins.subscriptions s
229 ON u.user_id = s.user_id;
230
231
232
```

Add parameter

Table	+	Q	Y	
	1 ² ₃ user_id	A ^B _C user_name	1 ² ₃ subscription_id	📅 start_date
1	1	Nomvula	501	2026-01-15
2	1	Nomvula	503	2026-03-10
3	3	Anele	505	2026-04-05
4	2	David	502	2026-02-01
5	4	Kabelo	null	null
6	5	Lerato	null	null
7	null	null	504	2026-03-20

Question 12

```
231  --Question 12:Show every plan and every subscription, including plans without subscribers AND any subscription
232  --referencing a plan that does not exist.
233  SELECT DISTINCT
234    p.plan_id,
235    p.plan_name,
236    s.subscription_id,
237    s.user_id
238  FROM brightlearn.sql_joins.plans p
239  FULL OUTER JOIN brightlearn.sql_joins.subscriptions s
240  ON p.plan_id = s.plan_id;
```

Add parameter

Table +					🔍 🔧 📄 📊 📄				
	¹ ₃ plan_id	^A ₀ plan_name	¹ ₃ subscription_id	¹ ₃ user_id					
1	11	Standard	504	6					
2	11	Standard	502	2					
3	12	Premium	503	1					
4	10	Basic	501	1					
5	13	Family	505	3					
6	14	Mobile	null	null					

Question 13

```
241 --Question 13:Show every show and every viewing session, including shows that were never watched
242 --AND sessions referencing shows that do not exist.
243 SELECT DISTINCT
244     s.show_id,
245     s.show_title,
246     v.session_id,
247     v.watch_minutes
248 FROM brightlearn.sql_joins.shows s
249 FULL OUTER JOIN brightlearn.sql_joins.viewing_sessions v
250 ON s.show_id = v.show_id;
251
252
```

Add parameter

Table ▾ +



	¹ ₃ show_id	^A _C show_title	¹ ₃ session_id	¹ ₃ watch_minutes
1	704	Cooking Lab	null	null
2	701	Comedy Hour	904	20
3	703	Tech Tales	902	30
4	705	Wild Earth	905	90
5	702	Crime Time	903	60
6	706	Wild Earth	null	null
7	701	Comedy Hour	901	45

Question 14

```
252 --Question 14:Show every user and every viewing session, including users with no sessions AND
253 --sessions referencing users who do not exist
254 SELECT DISTINCT
255     u.user_id,
256     u.user_name,
257     v.session_id,
258     v.show_id,
259     v.watch_minutes
260 FROM brightlearn.sql_joins.users u
261 FULL OUTER JOIN brightlearn.sql_joins.viewing_sessions v
262 ON u.user_id = v.user_id;
263
264
```

Add parameter

Table ▾ +					
	^{1 2 3} user_id	^{A B C} user_name	^{1 2 3} session_id	^{1 2 3} show_id	^{1 2 3} watch_minutes
1	null	null	904	701	20
2	1	Nomvula	903	702	60
3	3	Anele	905	705	90
4	1	Nomvula	901	701	45
5	4	Kabelo	null	null	null
6	5	Lerato	null	null	null
7	2	David	902	703	30

Question 15

```

264 --Question 15:Show every user, every subscription, and every plan in one query – using FULL OUTER
265 --JOIN throughout. This is the hardest question – get all gaps visible at once.
266 SELECT DISTINCT
267     u.user_id,
268     u.user_name,
269     s.subscription_id,
270     p.plan_id,
271     p.plan_name
272 FROM brightlearn.sql_joins.users u
273 FULL OUTER JOIN brightlearn.sql_joins.subscriptions s
274 ON u.user_id = s.user_id
275 FULL OUTER JOIN brightlearn.sql_joins.plans p
276 ON s.plan_id = p.plan_id;
277

```

Add parameter

Table	+					
	¹ ₃ user_id	^A _C user_name	¹ ₃ subscription_id	¹ ₃ plan_id	^A _C plan_name	
2	3	Anele	505	13	Family	
3	2	David	502	11	Standard	
4	4	Kabelo	null	null	null	
5	5	Lerato	null	null	null	
6	null	null	null	14	Mobile	
7	1	Nomvula	503	12	Premium	
8	1	Nomvula	501	10	Basic	